

```
-- =====
-- Crea la base de datos y las tablas
-- =====
```

```
CREATE DATABASE alkewallet;
```

```
USE alkewallet;
```

```
CREATE TABLE moneda(
currency_id INT AUTO_INCREMENT PRIMARY KEY,
currency_name VARCHAR(50) NOT NULL UNIQUE,
currency_symbol VARCHAR(5) NOT NULL UNIQUE
);
```

```
CREATE TABLE usuarios(
user_id INT AUTO_INCREMENT PRIMARY KEY,
nombre VARCHAR(30) NOT NULL,
apellido VARCHAR(30) NOT NULL,
correo VARCHAR(50) NOT NULL UNIQUE,
clave VARCHAR(30) NOT NULL,
saldo DECIMAL(30,2) NOT NULL CHECK (saldo >= 0),
currency_id INT,
```

```
FOREIGN KEY(currency_id) REFERENCES moneda(currency_id) ON DELETE SET NULL
);
```

```
CREATE TABLE transacciones(
transaction_id INT AUTO_INCREMENT PRIMARY KEY,
sender_user_id INT NOT NULL,
receiver_user_id INT NOT NULL,
importe DECIMAL(10,2) NOT NULL,
transaction_date TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
currency_id INT NOT NULL,
```

```
FOREIGN KEY (sender_user_id) REFERENCES usuarios(user_id),
FOREIGN KEY(receiver_user_id) REFERENCES usuarios(user_id),
FOREIGN KEY(currency_id) REFERENCES moneda(currency_id)
);
```

	#	Time	Action	Message
✓	1	17:25:15	CREATE DATABASE alkewallet	1 row(s) affected
✓	2	17:25:15	USE alkewallet	0 row(s) affected
✓	3	17:25:15	CREATE TABLE moneda(currency_id INT AUTO_INCREMENT PRIMARY KEY, currency_name VARCHA...	0 row(s) affected
✓	4	17:25:15	CREATE TABLE usuarios(user_id INT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(30) NOT ...	0 row(s) affected
✓	5	17:25:15	CREATE TABLE transacciones(transaction_id INT AUTO_INCREMENT PRIMARY KEY, sender_user_id I...	0 row(s) affected

```
INSERT INTO usuarios(nombre,apellido,correo,clave,saldo,currency_id)
VALUES
("Juan","Pérez","juan_perez@gmail.com","1234",2500.33,2),
("María","Dolores","maria_dolores@gmail.com","432112,4",100000,2),
("Mario","Bros","mario_bros@gmail.com","88224466ab",60000.55,3),
("Hugo","Patiño","hugo_patiño@gmail.com","qwerty",55555,1),
("Paco","Patiño","paco_patiño@gmail.com","micontraseña123",8888.55,3),
("Luis","Patiño","luis_patiño@gmail.com","admin",888888.55,3);
```

```
INSERT INTO transacciones(sender_user_id,receiver_user_id,importe,currency_id)
VALUES
(1,4,55555,3),
(3,2,456489,1),
(6,5,1000,2),
(5,4,1000,3),
(4,6,1000,1);
```

[illegible]

```
-- =====
-- CONSULTAS
-- =====
```

-- 1) Nombre de la moneda elegida por un usuario específico.

```
SELECT CONCAT(u.nombre," ",u.apellido) as Usuario, m.currency_name as "Moneda
Favorita"
FROM usuarios u
JOIN moneda m ON
u.currency_id = m.currency_id
where u.user_id = 3;
```

	Usuario	Moneda Favorita
▶	Mario Bros	Yuan

-- 2) Todas las transacciones registradas.

```
SELECT *
FROM transacciones;
```

	transaction_id	sender_user_id	receiver_user_id	importe	transaction_date	currency_id
▶	1	1	4	55555.00	2026-07-04 17:29:34	3
	2	3	2	456489.00	2026-07-04 17:29:34	1
	3	6	5	1000.00	2026-07-04 17:29:34	2
	4	5	4	1000.00	2026-07-04 17:29:34	3
	5	4	6	1000.00	2026-07-04 17:29:34	1
*	NULL	NULL	NULL	NULL	NULL	NULL

-- o también de manera más visual...

```
SELECT
t.transaction_id,
CONCAT(sender.nombre," ",sender.apellido," (",t.sender_user_id,")") as Sender,
CONCAT(receiver.nombre," ",receiver.apellido," (",t.receiver_user_id,")") as Receiver,
CONCAT(m.currency_symbol,t.importe) as Importe,
m.currency_name as Moneda,
t.transaction_date
```

FROM transacciones as t

```
JOIN usuarios sender ON
t.sender_user_id = sender.user_id
JOIN usuarios receiver ON
t.receiver_user_id = receiver.user_id
JOIN moneda m ON
t.currency_id = m.currency_id;
```

	transaction_id	Sender	Receiver	Importe	Moneda	transaction_date
▶	1	Juan Pérez (1)	Hugo Patiño (4)	¥55555.00	Yuan	2026-07-04 17:29:34
	2	Mario Bros (3)	María Dolores (2)	\$456489.00	Peso Chileno	2026-07-04 17:29:34
	3	Luis Patiño (6)	Paco Patiño (5)	USD\$1000.00	Dólar	2026-07-04 17:29:34
	4	Paco Patiño (5)	Hugo Patiño (4)	¥1000.00	Yuan	2026-07-04 17:29:34
	5	Hugo Patiño (4)	Luis Patiño (6)	\$1000.00	Peso Chileno	2026-07-04 17:29:34

-- 3) Todas las transacciones realizadas por un usuario específico

```
SELECT *
FROM transacciones
WHERE sender_user_id = 4 or receiver_user_id = 4;
```

	transaction_id	sender_user_id	receiver_user_id	importe	transaction_date	currency_id
▶	1	1	4	55555.00	2026-07-04 17:29:34	3
	4	5	4	1000.00	2026-07-04 17:29:34	3
	5	4	6	1000.00	2026-07-04 17:29:34	1
*	NULL	NULL	NULL	NULL	NULL	NULL

-- o al igual que en el ejemplo anterior, de manera más visual...

```
SELECT
t.transaction_id,
CONCAT(sender.nombre, " ", sender.apellido, " (", t.sender_user_id, ")") as Sender,
CONCAT(receiver.nombre, " ", receiver.apellido, " (", t.receiver_user_id, ")") as Receiver,
CONCAT(m.currency_symbol, t.importe) as Importe,
m.currency_name as Moneda,
t.transaction_date
```

FROM transacciones as t

JOIN usuarios sender ON

t.sender_user_id = sender.user_id

JOIN usuarios receiver ON

t.receiver_user_id = receiver.user_id

JOIN moneda m ON

t.currency_id = m.currency_id

WHERE t.sender_user_id = 4 or t.receiver_user_id = 4;

	transaction_id	Sender	Receiver	Importe	Moneda	transaction_date
▶	1	Juan Pérez (1)	Hugo Patiño (4)	¥55555.00	Yuan	2026-07-04 17:29:34
	4	Paco Patiño (5)	Hugo Patiño (4)	¥1000.00	Yuan	2026-07-04 17:29:34
	5	Hugo Patiño (4)	Luis Patiño (6)	\$1000.00	Peso Chileno	2026-07-04 17:29:34

-- 4) DML: modificar el correo electrónico de un usuario específico.

UPDATE usuarios

SET correo = "juan_perez_123456@gmail.com"

WHERE user_id = 1;

SELECT nombre, correo

FROM usuarios

WHERE user_id = 1;

	nombre	correo
▶	Juan	juan_perez@gmail.com

--Luego-->

	nombre	correo
▶	Juan	juan_perez_123456@gmail.com

-- 5) DML: eliminar los datos de una transaccion (fila completa).

```
DELETE
FROM transacciones
WHERE transaction_id = 1;
```

```
SELECT *
FROM transacciones;
```

– Primero

	transaction_id	sender_user_id	receiver_user_id	importe	transaction_date	currency_id
▶	1	1	4	55555.00	2026-07-04 17:40:46	3
	2	3	2	456489.00	2026-07-04 17:40:46	1
	3	6	5	1000.00	2026-07-04 17:40:46	2
	4	5	4	1000.00	2026-07-04 17:40:46	3
	5	4	6	1000.00	2026-07-04 17:40:46	1
*	NULL	NULL	NULL	NULL	NULL	NULL

– Luego

	transaction_id	sender_user_id	receiver_user_id	importe	transaction_date	currency_id
▶	2	3	2	456489.00	2026-07-04 17:29:34	1
	3	6	5	1000.00	2026-07-04 17:29:34	2
	4	5	4	1000.00	2026-07-04 17:29:34	3
	5	4	6	1000.00	2026-07-04 17:29:34	1
*	NULL	NULL	NULL	NULL	NULL	NULL